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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

IV Semester B. E. Regular Examinations Sep/Oct – 2024

Industrial Engineering and Management

OPERATIONS RESEARCH

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What is the difference between a descriptive and a prescriptive models in operations research?	02	2	2
	1.2	List the major assumptions of linear programming.	02	2	2
	1.3	In simplex method, what is the purpose of the pivot operation?	02	3	3
	1.4	When can the dual Simplex method be applied?	02	3	3
	1.5	List the methods used to arrive at an initial basic feasible solution in transportation model.	02	2	2
	1.6	How does a travelling salesman problem differ from a routine assignment model?	02	3	3
	1.7	If an activity has its optimistic, most likely and pessimistic times as $2, 4$ *and* 9 respectively, then what is its expected time and variance?	02	3	3
	1.8	Which project management tool introduces the estimates for optimistic time and pessimistic time?	02	2	2
	1.9	Define pay –off as applied to game theory.	02	2	2
	1.10	What is a two person zero sum game?	02	2	2

PART-B

2	a	List the seven step model building process with an example for each step.	04	2	2
	b	An Electric Company produces two products P_1 and P_2. Products are produced and sold on a weekly basis. The weekly production cannot exceed 25 for product P_1 and 35 for product P_2 because of limited available facilities. The company employs total of 60 workers. Product P_1 requires 2 man-weeks of labor, while P_2 requiems one man-week of labour. Profit margin on P_1 is Rs. 60 and P_2 is Rs. 40. Formulate this problem as an LP problem and solve that using graphical method.	12	4	4
3	a	Discuss the following with respect to linear program: i) Slack variable ii) Surplus variable	04	2	2
	b	Solve the following linear programming problem by simplex method: $Maximize\ Z = 3x_1 + 5x_2 + 4x_3$ Subject to the constraints $2x_1 + 3x_2 \leq 8$ $2x_2 + 5x_3 \leq 10$ $3x_1 + 2x_2 + 4x_3 \leq 15$ $and\ x_1, x_2, x_3 \geq 0.$	12	3	3

OR

4 a Enumerate important characteristics of Duality. 06 2 2

b Solve by dual simplex method the following linear programming problem:

Minimize $Z = 6x_1 + 7x_2 + 3x_3 + 5x_4$

Subject to

$5x_1 + 6x_2 - 3x_3 + 4x_4 \geq 12$

$x_2 + 5x_3 - 6x_4 \geq 10$

$2x_1 + 5x_2 + x_3 + x_4 \geq 8$

$x_1, x_2, x_3, x_4 \geq 0$ 10 3 3

5 a Solve the initial basic feasible solution of transportation problem by least cost entry method:

		Destination			Supply
		Kanpur	Pune	Delhi	
Source	Jaipur	4	5	1	40
	Udaipur	3	4	3	60
	Mumbai	6	2	8	70
Demand		70	40	60	

04 2 2

b Find the optimum solution to the following transportation problem in which the cells contain the transportation cost in rupees.

	W_1	W_2	W_3	W_4	W_5	Available
F_1	7	6	4	5	9	40
F_2	8	5	6	7	8	30
F_3	6	8	9	6	5	20
F_4	5	7	7	8	6	10
Required	30	30	15	20	5	

12 3 3

OR

6 a Four different jobs can be done at four different machines. The set up and take down time costs are assumed to be prohibitively high for changeovers. The matrix below gives the cost in rupees of producing job i on machine j.

		Machines			
		M_1	M_2	W_3	M_4
Jobs	J_1	5	7	11	6
	J_2	8	5	9	6
	J_3	4	7	10	7
	J_4	10	4	8	3

How should the jobs be assigned to the various machines so that the total cost is minimized. 08 3 3

b ABC Ice cream company has a distribution depot in Greater Kailash Part 1 for distributing ice-cream in South Delhi. There are four vendors located in different parts of South Delhi (call them A, B, C *and* D) who have to be supplied ice-cream every day. The following matrix displays the distances (in kilometers) between the depot and the four vendors:

		To				
		Depot	*Vendor A*	*Vendor B*	*Vendor C*	*Vendor D*
	Depot	∞	3.5	3	4	2
	Vendor A	3.5	∞	4	2.5	3
Form	*Vendor B*	3	4	∞	4.5	3.5
	Vendor C	4	2.5	4.5	∞	4
	Vendor D	2	3	3.5	4	∞

What route should the company van follow so that the total distance travelled is minimized? 08 3 3

Q		Question	Marks		
7	a	List the steps for drawing a network.	08	2	2
	b	A project schedule has the following characteristics: (see table below) i) Construct the network ii) Compute E_i and L_j values for each network iii) Find the critical path and total duration	08	3	3

Activity	*Time(weeks)*	*Activity*	*Tim (weeks)*
1 – 2	4	5 – 6	4
1 – 3	1	5 – 7	8
2 – 4	1	6 – 8	1
3 – 4	1	7 – 8	2
3 – 5	6	8 – 10	5
4 – 9	5	9 – 10	7

OR

Q		Question	Marks		
8	a	What is float? What are the different types of floats?	04	2	2
	b	A project schedule has the following characteristics: (see table below) i) Calculate the expected time for each activity ii) Draw the network diagram iii) Find the critical path and duration iv) Calculate total float for each activity	12	3	3

Activity	10 – 20	20 – 30	20 – 40	20 – 50	30 – 60	40 – 60	50 – 70	60 – 70
Optimistic	1	8	3	3	4	7	6	1
Most likely	2	10	4	5	5	8	8	4
Pessimistic	3	12	6	7	6	9	10	7

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9 a A company management and the labor union are negotiating a new three – year settlement. Each of these has 4 strategies:

i) Hard and aggressive bargaining,
ii) Reasoning and logical approach,
iii) Legalistic strategy
iv) Conciliatory approach. The costs to the company are given for every pair of strategy choice.

Company strategies

Union Strategies	*I*	*II*	*III*	*IV*
I	20	15	12	35
II	25	14	8	10
III	40	2	10	5
IV	−5	4	11	0

Find the saddle point and value of the game. — 06 | 3 | 3

b Use the graphical method for solving the following game and find the value of the game.

Player B

Player A	B_1	B_2	B_3	B_4
A_1	2	2	3	-2
A_2	4	3	2	6

— 10 | 3 | 3

OR

10 a List the assumption of a game theory. — 08 | 2 | 2

b Reduce the following game by dominance and find the optimal strategies and value of the game.

Plaver B

Player A	*I*	*II*	*III*	*IV*
I	3	2	4	0
II	3	4	2	4
III	4	2	4	0
IV	0	4	0	8

— 08 | 3 | 3

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R V College of Engineering

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IV Semester BE Regular/Supplementary Examinations June/July - 2025.
Course : Operations Research-IM244AI

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of normal distribution table is permitted.

Part A

Question No	Question	M	CO	BT
1.1	Operations research is a .... approach to problem solving for executives.	01	1	2
1.2	Any two isoprofit or isocost lines for a given linear programming problem are .... to each other.	01	2	1
1.3	The constraints may be in the form of .... or	02	2	1
1.4	Define sensitivity analysis in linear programming.	02	1	1
1.5	What does the shadow price in duality represent?	02	2	2
1.6	What is the goal in a transportation problem?	01	2	1
1.7	What is the cost matrix in assignment problems?	01	2	2
1.8	What are the steps of the Hungarian method?	02	3	2
1.9	What is the full form of CPM?	01	1	1
1.10	What is a critical path?	01	2	2

1.11	What are the advantages of crashing a network?	02	3	2
1.12	Define saddle point in a game matrix.	02	2	1
1.13	When is a game solved using the arithmetic method?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	With an example, explain a work-scheduling problem in linear programming.	06	2	3
2b	Explain the concept of feasible region and optimal solution in a linear programming problem using an example.	06	2	3
2c	Define Operations Research and mention any four of its applications.	04	1	1
3a	Use penalty (Big-M) method to solve the following LP problem. Minimize Z = 5x1 + 3x2 subject to the constraints (i) 2x1 + 4x2 ≤ 12, (ii) 2x1 + 2x2 = 10, (iii) 5x1 + 2x2 ≥ 10 and x1, x2 ≥ 0.	08	2	3
3b	Formulate the **dual** of the following problem and interpret the dual variables economically: Maximize Z=6x+8y s.t. 2x+3y≤30, x+2y≤20, x,y≥0	08	2	3
	OR			
4a	Write the standard format of simplex method for the following LP problem. Maximize Z = 3x1 + 5x2 + 4x3 subject to the constraints (i) 2x1 + 3x2 ≤ 8, (ii) 2x2 + 5x3 ≤ 10, (iii) 3x1 + 2x2 + 4x3 ≤ 15 and x1, x2, x3 ≥ 0	06	2	4
4b	Conduct **sensitivity analysis** for the objective coefficients of the LP: Maximize Z=2x+3y s.t. x+y≤4 x+2y≤6 x,y≥0 after obtaining optimal solution.	10	2	4
5a	A company wants to assign 4 salesmen (S1, S2, S3, S4) to 4 different sales regions (R1, R2, R3, R4). The expected profits (in ₹'000s) from	10	3	3

each salesman in each region are given in the table below. Assign the salesmen to the regions such that the total profit is maximized.

	R1	R2	R3	R4
S1	25	40	35	20
S2	30	45	40	25
S3	20	35	30	50
S4	10	25	20	30

5b	Compare the three methods of finding initial basic feasible solution: North West Corner, Least Cost, Vogel's Method.	06	2	4

OR

6a	Discuss degeneracy in a transportation problem. How is it identified and resolved?	06	3	2
6b	Explain the general structure and assumptions of a Transportation Problem.	06	1	2
6c	Explain any two methods to find the initial basic feasible solution.	04	2	2

7a — A project consists of the following activities with the duration in days: — 10 | 2 | 4

Activity	A	B	C	D	E	F	G	H	I
Precedence	-	A	A	B,C	A	D,E	C	F,G	H
Duration	10	4	12	10	9	8	7	9	6

i. Draw the network of the above project
ii. Identify the critical path and determine the project duration
iii. Calculate independent float, free float for all the activities.

7b — An assembly is to be made from two parts X and Y. Both parts must be turned on a lathe. Y must be polished whereas X need not be polished. The sequence of activities, together with their predecessors, is given below. — 06 | 3 | 4

Activity	Description	Predecessor Activity
A	Open work order	–
B	Get material for X	A
C	Get material for Y	A
D	Turn X on lathe	B
E	Turn Y on lathe	B, C
F	Polish Y	E
G	Assemble X and Y	D, F
H	Pack	G

Draw a network diagram of activities for the project

OR

8a	'PERT takes care of uncertain durations.' How far is this statement correct? Explain with reasons.	08	3	4
8b	Explain the following in the context of project management: (i) Activity variance (ii) Project variance	08	3	2
9a	Explain: Minimax and Maximin principle used in the theory of games.	06	4	2
9b	By applying dominance rules, reduce the given payoff matrix and identify the optimal pure strategies and value of the game.	10	4	4

	B1	B2	B3	B4
A1	1	2	1	3
A2	0	-1	0	2
A3	2	3	1	4

OR

10a	Obtain the optimal strategies for both persons and the value of the game for two-person zero-sum game whose payoff matrix is as follows:	10	4	3

Player B

Player A	B_1	B_2
A_1	1	−3
A_2	3	5
A_3	−1	6
A_4	4	1
A_5	2	2
A_6	−5	0

10b	Game theory provides a systematic quantitative approach for analysing competitive situations in which the competitors make use of logical processes and techniques in order to determine an optimal strategy for winning. Comment.	06	4	5